Automated Pick-and Place System

Project Description

As part of this course, you will develop a working solution to an industrial automation problem using the theory and methods introduced in the lectures. You are not required to write a full report describing your solution. However, the solution you build will form the **basis for your oral exam**, where it you are expected to present your work in a well-structured manner.

Project Setup

You will work in **six groups** of approximately equal size. Each group will be assigned the same automation challenge:

Automation Task

Design, configure, and implement a system that:

- Detects components on a moving conveyor belt
- Uses an articulated robot to pick up these components from their measured location
- Places the components in predefined, fixed locations

Your solution must ensure safe, reliable, and repeatable operation by:

1. Configuring the PLC and conveyor system (frequency converter)
2. Programming the logic required for detection, pick-and-place operations, and system control
3. Programming the robot to execute pick and place of the components
4. Ensuring that all components and program logic work correctly through the provided digital I/O interfaces
5. Design a part feeding system to dispense parts to the conveyor
6. Design fingers for the gripper that fits the part to be picked
7. Implement a OPCUA server on the PLC to enable reading and writing from a PC to:
    1. Start/stop
    2. Set conveyor speed
    3. Readout productions statistics
8. Evaluate system performance, safety, system robustness, and practical operation

Available Hardware

Each group will work with a provided hardware setup consisting of:

- KUKA 6 axis industrial robot (actual names here)
- Pneumatic gripper system
- Conveyor belt and frequency converter
- Part detection sensor
- PLC for centralized system control
- Digital I/O communication between the PLC and the robot
- Safety system, including:
    - A light curtain
    - A safety relay

Some parts of the setup will already be preconfigured due to hardware limitations. It is important that you:

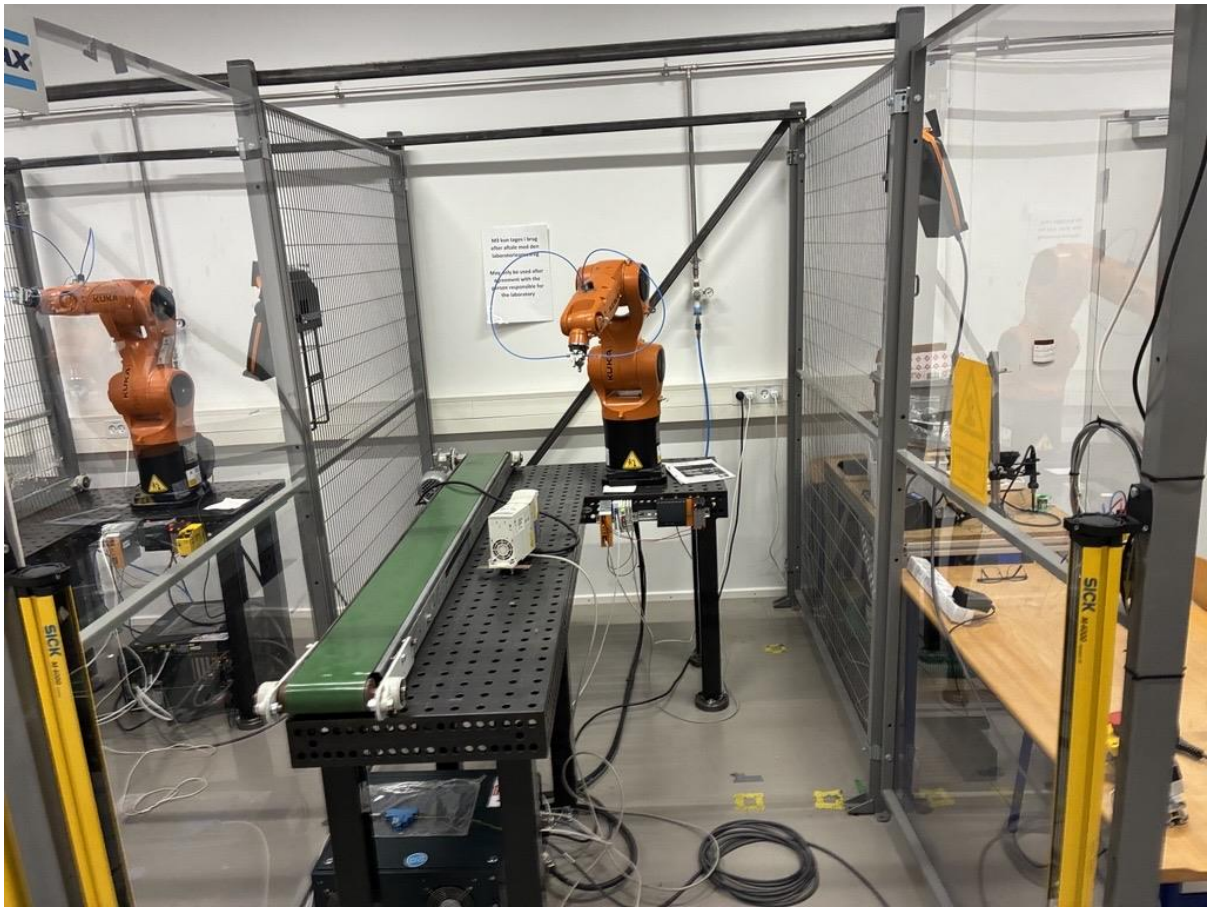
- Study these preconfigured components
- Understand why they are implemented as they are
- Reflect on the effectiveness of the implementation and whether they could be improved

Outcome

By the end of the project, your group should have a functioning automated pick-and-place system. You should be prepared to explain:

- The system architecture
- The programming decisions
- How the components interact
- How safety has been ensured
- Any limitations or potential improvements
- The weight limit of the Conveyor

This explanation will be part of your **oral exam**, so make sure you understand the full solution—not just the parts you personally implemented.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